Nadir and Wide Swath Altimetry for Inland Water Bodies

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DOCUMENT CONTROL SHEET

1.	Security Classification	Non-classified			
2	Distribution	General			
3.	Document version	1			
4.	Document Type	Surface Water and Ocean Topography (SWOT) Report			
5.	Document Control Number				
	Title	Nadir and Wide Swath Altimetry for Inland Water Bodies			
7.	Particular of collation	Pages 25	Figures 19	Tables 0	References 20
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11.	Originating unit	IIRS, Dehradun			
12.	Sponser(s)/ Name and Address	Indian Space Research Organisation, Department of Space, Govt. of India			
13.	Date of Initiation	28 April 2025			
14.	Date of Publication	14 May 2025			
15.	Abstract (with keywords)	Satellite altimetry has revolutionized the monitoring of inland water bodies, providing essential data for			

hydrology, water resource management, and climate studies. The Indian Institute of Remote Sensing (IIRS) has been at the forefront of utilizing space-based altimetry for inland water level monitoring, addressing the challenges of sparse ground-based observations and the need for consistent, global data.

Early missions like Topex/Poseidon, ERS-2, Jason-1, and Envisat-RA2 demonstrated the potential of altimetry, providing inland water level data. Despite these advancements, traditional nadir altimeters, which measure directly beneath the satellite, have limitations in spatial coverage and resolution. To overcome these gaps, wide-swath altimetry missions have been developed.

The Surface Water and Ocean Topography (SWOT) mission represents a significant leap forward. It employs wide-swath interferometric altimetry using the KaRIn instrument to map water surface elevations across broad areas, not just along the satellite's ground track. This enables high-resolution, two-dimensional mapping of lakes, reservoirs, wetlands, and rivers wider than 100 meters. Case studies presented in the report demonstrate the use of radar altimetry for generating rating curves in the Mahanadi River, Odisha and monitoring water levels in Indian reservoirs. SWOT's wide-swath data has been instrumental in observing changes in water bodies and managing hydropower projects, illustrating its transformative potential for water management.

In summary, space-based altimetry, especially with the advent of wide-swath missions like SWOT, is transforming the monitoring and management of inland water bodies.

Keywords: Satellite Altimetry; Remote sensing; Inland Water Monitoring; SWOT Mission; Hydrodynamic modelling; River Discharge.

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1. Introduction

Oceans, lakes, and rivers are major components of Earth's surface water. Freshwater lakes and rivers are primary sources of water supply for drinking, agriculture, energy, and transportation, and support aquatic ecosystems and wildlife. [1]

On land, most water accessible to humans is stored in lakes, rivers, soil and groundwater. Water enters these reservoirs through precipitation (rain, snow) and flow from their surrounding watersheds. Water leaves via evaporation, transpiration (evaporation through plant leaves) and river discharge into the ocean. A simplified view of the water cycle (Figure 1) shows a balance between water entering the land surface by precipitation and water leaving the land surface by evaporation, transpiration, and runoff. [2]

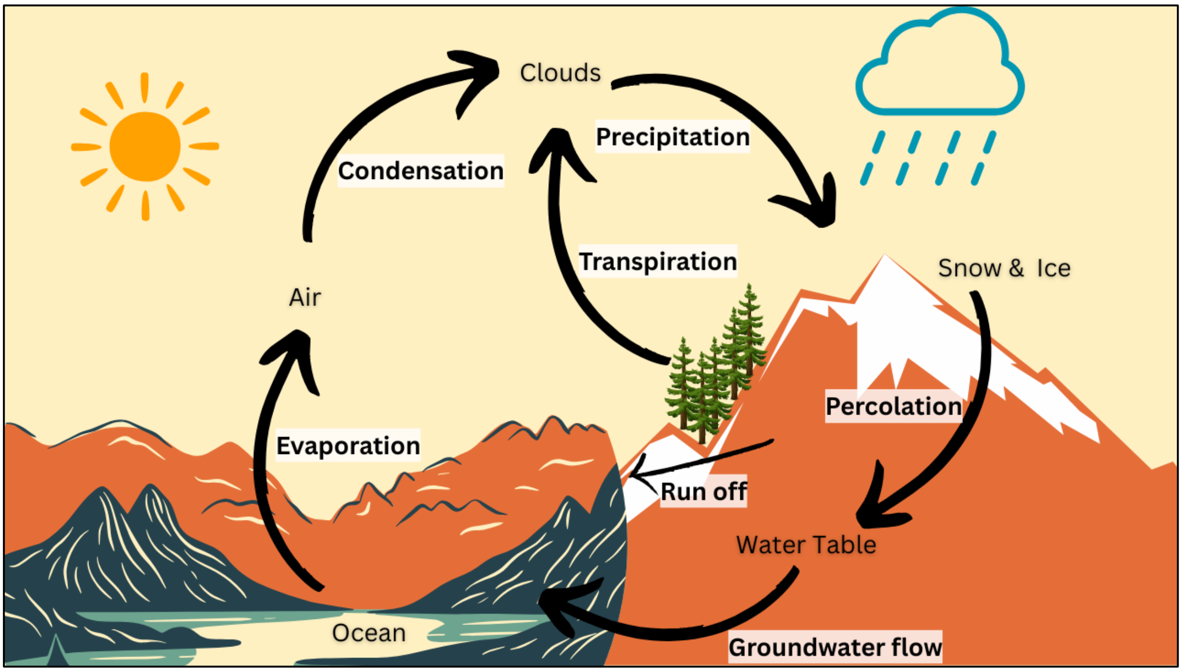

Figure 1. A simple Hydrological Cycle.

The inland water bodies contribute significantly to regional and global hydrological and biogeochemical cycles. Continuous monitoring of water levels is essential for understanding the global water cycle on land and study the dynamics of floodplains and wetlands, which influence flood control and the balance of ecosystems. Particularly, rivers spanning multiple countries or states within countries pose challenges in estimating water availability and usage.

Hydrologists use the stream gauge network, along with similarly vast networks of sensors measuring rainfall, soil moisture, snow depth and essential climate variables (ECVs). Understanding river flow, discharge, and the use of rating curves-which translate water level readings into streamflow estimates-is fundamental for effective river monitoring, hydrological analysis, and water resource management. [2]

The most sophisticated computer models include data assimilation, a process that continually compares previous forecasts with newly received measurements (e.g., actual rainfall data) to update and improve the model itself. But the current understanding of Earth's water balance on land is poor and this is partly because of a lack of global runoff data due to sparse ground-based observations. [2] But what if we could measure the water cycle from space?

For these inland waters, there are many applications under development and also an increasing range and sophistication of remote sensing capabilities. The challenge with applications and services related to inland waters is that the introduction of altimetry data requires a great deal of transdisciplinary knowledge. The recent development of on board DEM for current missions (e.g. Jason-3) as well the SAR technology (CryoSat-2, Sentinel-3) or LIDAR (ICESat, ICESat-2) help to provide better measurements and SWOT will be a major step forward to improve the global coverage and high resolution. [3]

2. Space-based Altimeters for Inland Water

2.1 Evolution of Altimetry

Early sea-farers on the trades routes were the first to gauge the velocity and direction of currents, such as in the Red Sea and the Mediterranean. To shorten the journey from New York to London, Benjamin Franklin recommended that captains carrying mail between America and Europe follow the Gulf Stream (Figure 2) to save time.

Figure 2. A schematic map of the Gulf Stream (https://scijinks.gov/gulf-stream/)

The satellite altimetry is initially designed to measure the sea level by a combination of radar technique (used to measure the distance from the satellite to a reflecting surface) and a positioning technique (allowing a very precise location of the satellite on its orbit).

In 1987, CNES and NASA formally created a partnership for a joint project involving French and American instruments, on an American satellite to be launched by a European rocket (Ariane 4): this was to become Topex/Poseidon.

Global Ocean observation is one of its assets: ten days of Topex/Poseidon satellite measurements provide more details than in-situ observations over the last hundred years. Figure 3 beautifully illustrates the Global Mean Sea Level Rise observed through different altimeters from 1993.

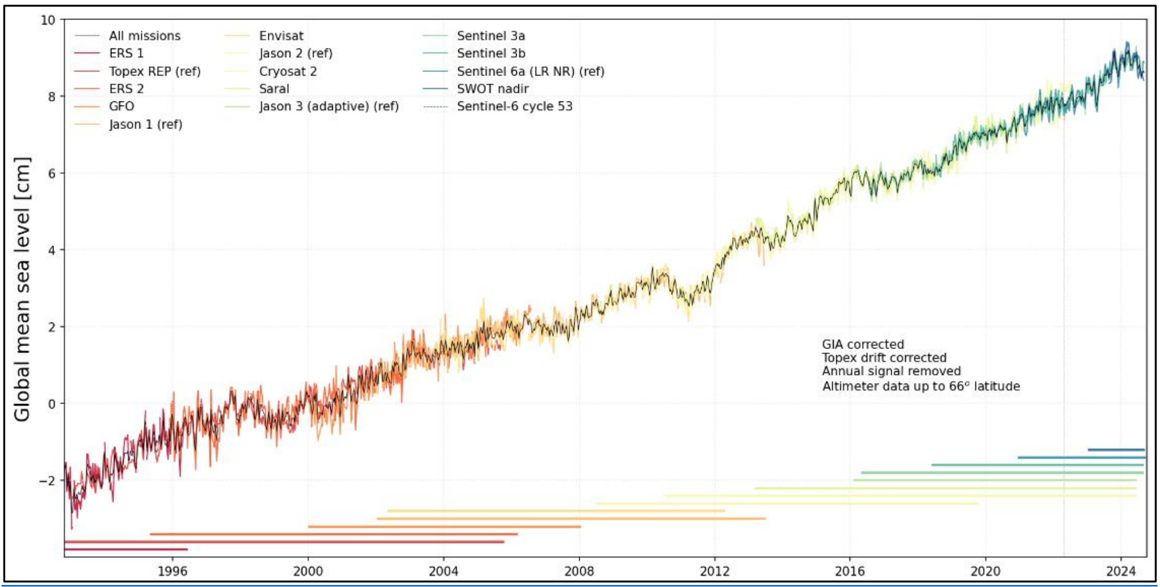

Figure 3. Global mean sea level between (GMSL) 66°N and 66°S based on 14 different satellites indicated, with respective colours, on the top and mission duration shown in the horizontal bars along the bottom. Credits EUMETSAT.

More and more faster processing systems allowed altimetry data in near real-time. Their assimilation into models, in combination with other data (in-situ, water temperature, salinity, etc.) has helped operational altimetry.

New results are obtained each day, the topics are diversifying, such as in hydrology on monitoring variations in lake levels (or rivers or wetlands), or in glaciology where altimetry brings its contribution to determine sea ice thickness or periods of freeze. [4]

2.2 Working Principle of Altimetry?

The time taken by a radar pulse to travel from the satellite antenna to the surface and back (radar echo), combined with precise satellite location data, is measured in altimetry. Altimetry satellites basically determine the distance from the satellite to a target surface by measuring the satellite-to-surface round-trip time of a radar pulse (Figure 4). However, this is not the only measurement made in the process, and a lot of other information can be extracted from altimetry, such as sea surface height, wave height, wind speed, ocean circulation patterns, as well as key parameters related to river dynamics, such as river water levels, flow variations, and surface slope.

The magnitude and shape of the echoes (or waveforms) also contain information about the characteristics of the surface which caused the reflection. The best results are obtained over the ocean, which is spatially homogeneous.

Basic Radar altimeters on board the satellite permanently transmit signals at high frequency (over 1700 pulses per second) to Earth, and receive the backscatter from the sea surface. This is analysed to derive a precise measurement of the round-trip time between the satellite and the sea surface. The time measurement, scaled by the speed of light, yields a range measurement. By averaging the estimates over a second, and after atmospheric corrections the final range R is estimated with an accuracy level upto 2 cm.

The ultimate aim is to measure sea level relative to a terrestrial reference frame. This requires independent measurements of the satellite orbital trajectory, i.e. exact latitude, longitude and altitude coordinates. [4]

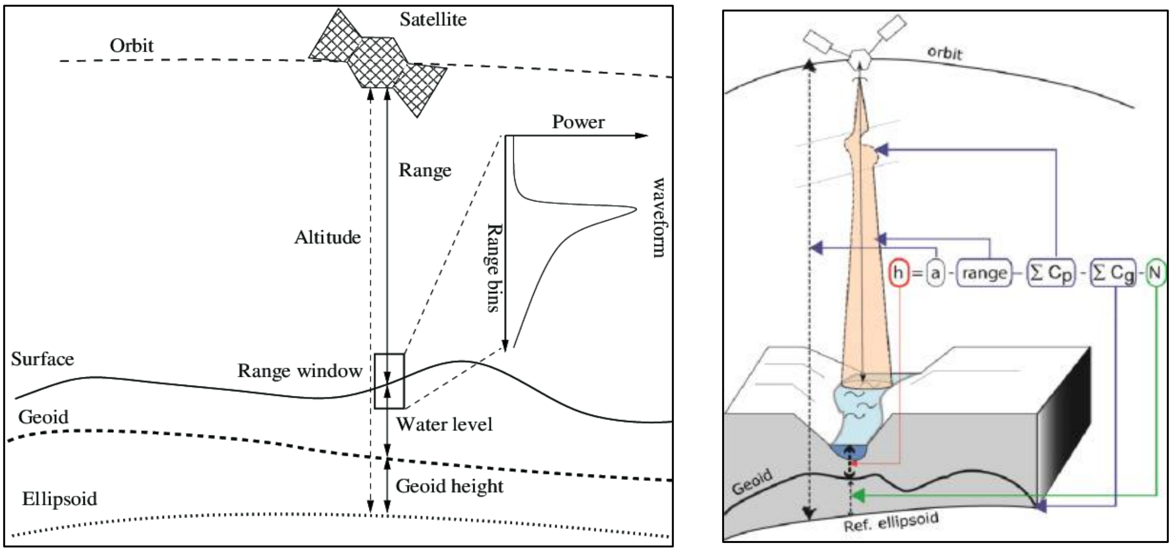

Figure 4. The principle of satellite altimetry [9] (left) Principles of altimeter measurement. The height of the reflective surface relative to a geodetic datum (ellipsoid) is given by the difference between the height of the satellite and the radar measurement, e.g. the range. Corrections must be made in order to consider both propagation in the atmosphere (C_p) and also vertical movements of the Earth's crust (C_g) and N is the local undulation of geoid. [13]

2.2.1 Frequencies Used

Several different frequencies are used for radar altimeters. The choice depends upon regulations, mission objectives and constraints, technical possibilities and impossibilities.

Ku band is the most commonly-used frequency (used for Topex/Poseidon, Jason-1, Envisat, ERS, etc). C and S bands are known to be more sensitive than Ku to ionospheric perturbation, and less sensitive to the effects of atmospheric liquid water. Its main function is to enable correction of the ionospheric delay in combination with the Ku-band measurements.

Signal frequencies in the Ka band enable better observation of ice, rain, coastal zones, land masses (forests, etc.) and wave heights. The first space borne altimeter to operate at Ka-band is SARAL/AltiKa. [4]

It still uses conventional pulse limited mode with innovative Ka band signals of which centre frequency is around 35.75 GHz. It has 500 MHz bandwidth so the pulse width is reduced to 2 ns and the effective footprint size is reduced to 1.3 km for calm water surface. AltiKa transmits on-ground at a rate of 40 Hz, which means a measurement every 175 m approximately. Provides much denser ground track than other altimetry missions which gives an opportunity to monitor a large number of waterbodies. The major drawback of SARAL is sensitive to rain and its 35-day repeat cycle which is barely sufficient correctly to monitor the hydrological cycle of the river. [15]

2.2.2 Waveforms and Retracking for heterogeneous surfaces

The magnitude and shape of the echoes (or waveforms) also contain information about the characteristics of the surface which caused the reflection. Over continental areas, the altimeter footprint is frequently contaminated by the multiplicity of surfaces. This contamination has important consequences on the complexity of the waveform (for example, many radar echoes are multi-peaked) and on the accuracy of measurements. The uniform sea surface and its perturbations receive backscatter in different forms, as is comprehensively demonstrated in Figure 6, helping us understand the waveform better and draw clearer insights from them.

Thus, all altimeter data for heterogeneous surfaces must be post-processed to produce accurate surface elevation measurements. This post-processing is called 'retracking' (Figure 5). The core formula for retracking involves identifying the leading edge of the waveform and then computing the retracked range by adding a correction based on the waveform's characteristic. [4]

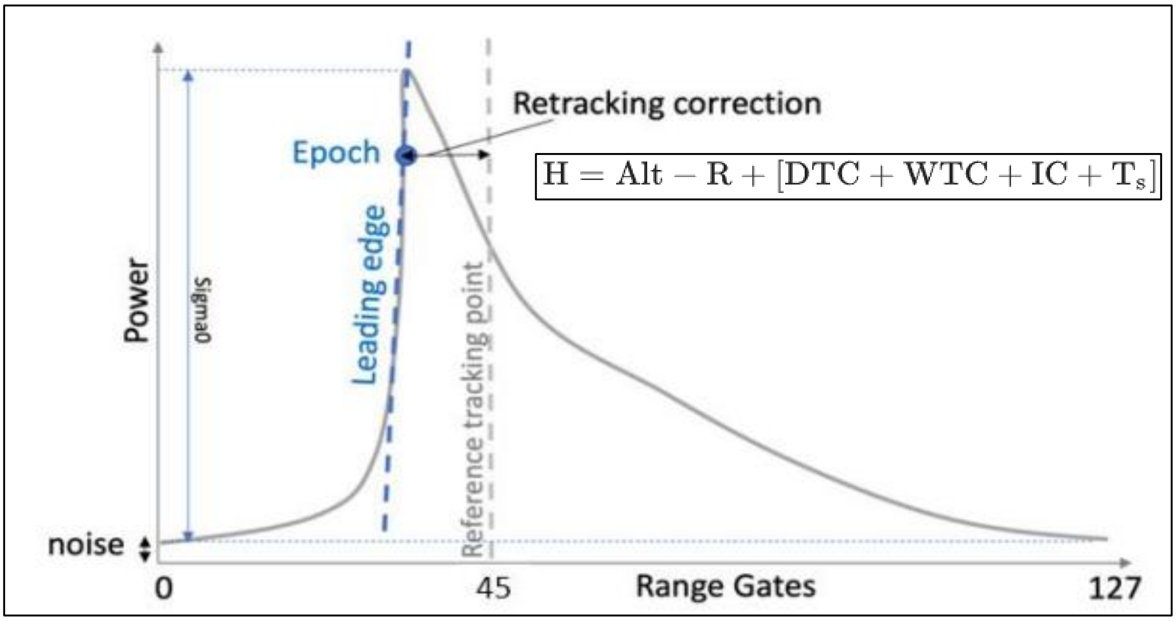

Figure 5. The height H of the reflecting surface is determined by the difference between the satellite orbit (Alt) and the altimeter range R measurement, with the addition of various corrections that take into account the time delays related to the propagation of the where DTC is the Dry Tropospheric Correction, WTC the Wet Tropospheric Correction, IC the Ionospheric Correction and T_s the solid tide correction. [13]

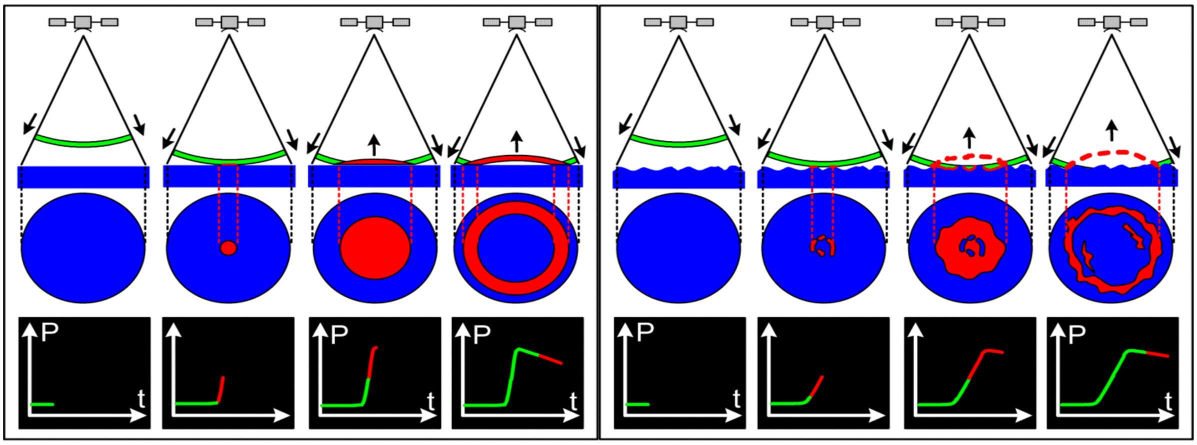

Figure 6. The radar altimeter receives the reflected wave (or echo), which varies in intensity over time. Where the sea surface is flat (left), the reflected wave's amplitude increases sharply from the moment the leading edge of the radar signal strikes the surface. However, in sea swell or rough seas (right), the wave strikes the crest of one wave and then a series of other crests which cause the reflected wave's amplitude to increase more gradually. We can derive ocean wave height from the information in this reflected wave, since the slope of the curve representing its amplitude over time is proportional to wave height. Credits CNES.

2.2.3 SAR Altimetry

The technique uses the same concept as Synthetic Aperture Radar (SAR) imagers by exploiting the Doppler effect caused by the satellite movement in the along-track direction. This provides a strong spatial resolution improvement in that direction, while across-track resolution remains the same than in conventional altimetry. E.g. Cryosat and the Sentinel-3A & B. [4]

Two main delay-Doppler processing have been developed for radar altimetry: the UnFocused SAR (UF-SAR) and Fully-Focused SAR (FF-SAR) processing (Figure 7).

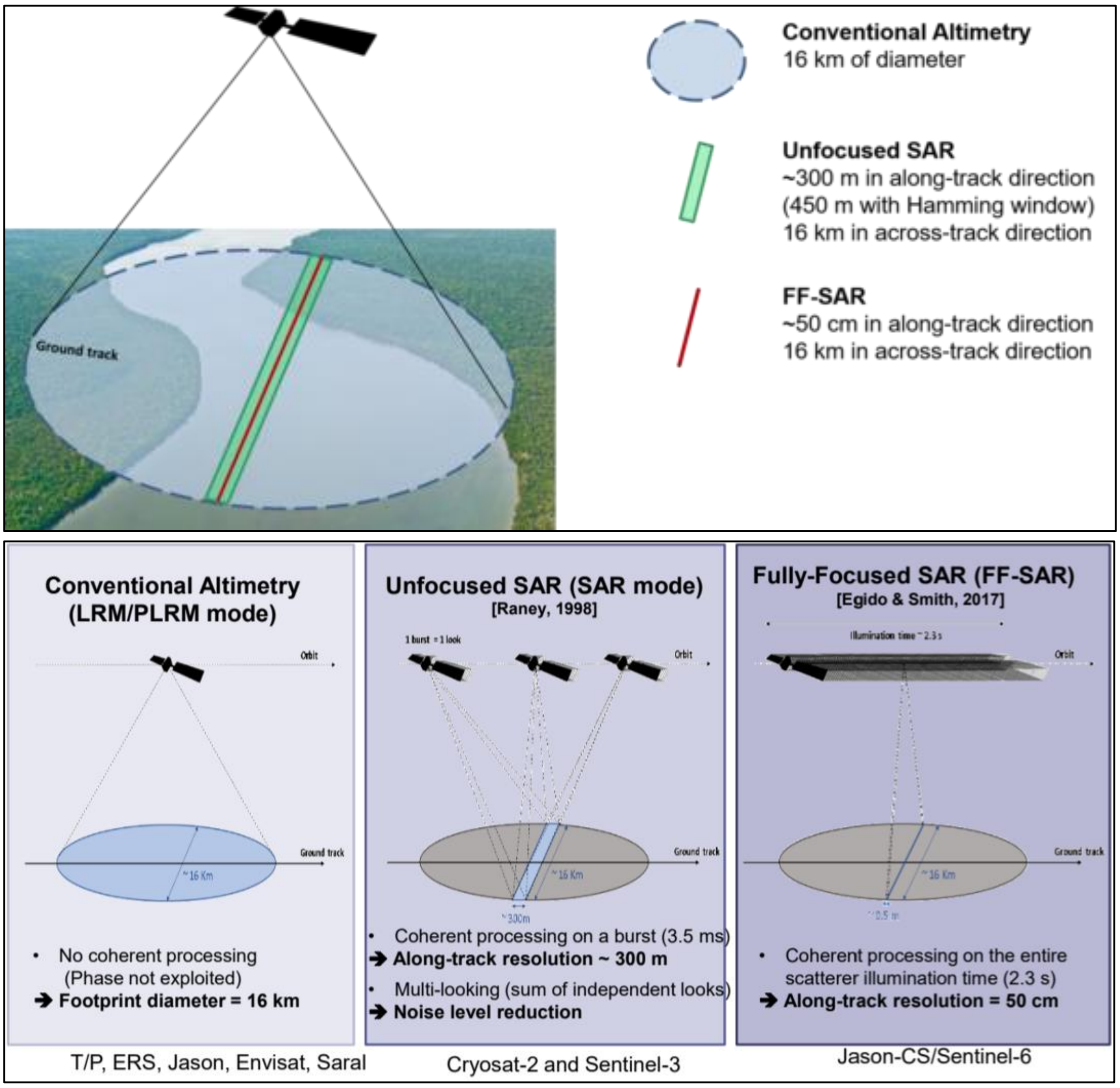

Figure 7. Modes/Techniques of SAR Altimetry. Credits AVISO.

2.2.4 Interferometric Altimetry

The altimetry interferometric technique relies on the measurement of the relative delay between observations of a same ground given point observed from two slightly different satellite positions. This relative delay expresses a slant range difference that can be interpreted as a geo-localised height measurement. [4]

The Surface Water and Ocean Topography (SWOT) mission marks a transformative leap in our ability to observe and understand Earth's water resources. First satellite with wide swath altimetry & provide 2D elevation of Earth's surface water-mapping oceans, lakes, and rivers. SWOT will scan the planet between 78°N latitude and 78°S latitude within 1 centimeter of accuracy and retracing the same path every 21 days.

This data will fill critical gaps in our knowledge of the water cycle, improve climate and weather models, and provide vital insights for managing freshwater resources and responding to extreme events such as floods and droughts. [20]

2.3 Gaps in Traditional Altimeters

A series of SAR systems for earth observations have been launched over the last three decades, with frequency bands ranging from X- to L-band, and incidence angles typically between 20 and 40° (10-50° in extreme cases).

These missions generally aimed at a relatively wide range of applications, and the main drivers for their evolution have been increased resolution, better spatiotemporal coverage, and improved polarimetric and interferometric acquisition capacities. More specialized SAR systems are now studied by several space agencies.

The main limitations of standard nadir-pointing radar altimeters have been understood for a long time. They include the lack of coverage (intertrack distance of typically 150 km for the T/P / Jason tandem) and the spatial resolution (typically 2 km for T/P and Jason) (Figure 8). [4]

Launched in December 2022, SWOT is the first satellite mission designed to provide a truly global survey of Earth's surface water, including detailed, repeated measurements of rivers, lakes, reservoirs, and the ocean. Using advanced radar interferometry, SWOT can observe ocean circulation at scales as fine as 15–25 km and resolve changes in water levels, stream slopes, and flooding extents in inland water bodies, offering a significant improvement over previous missions. [20]

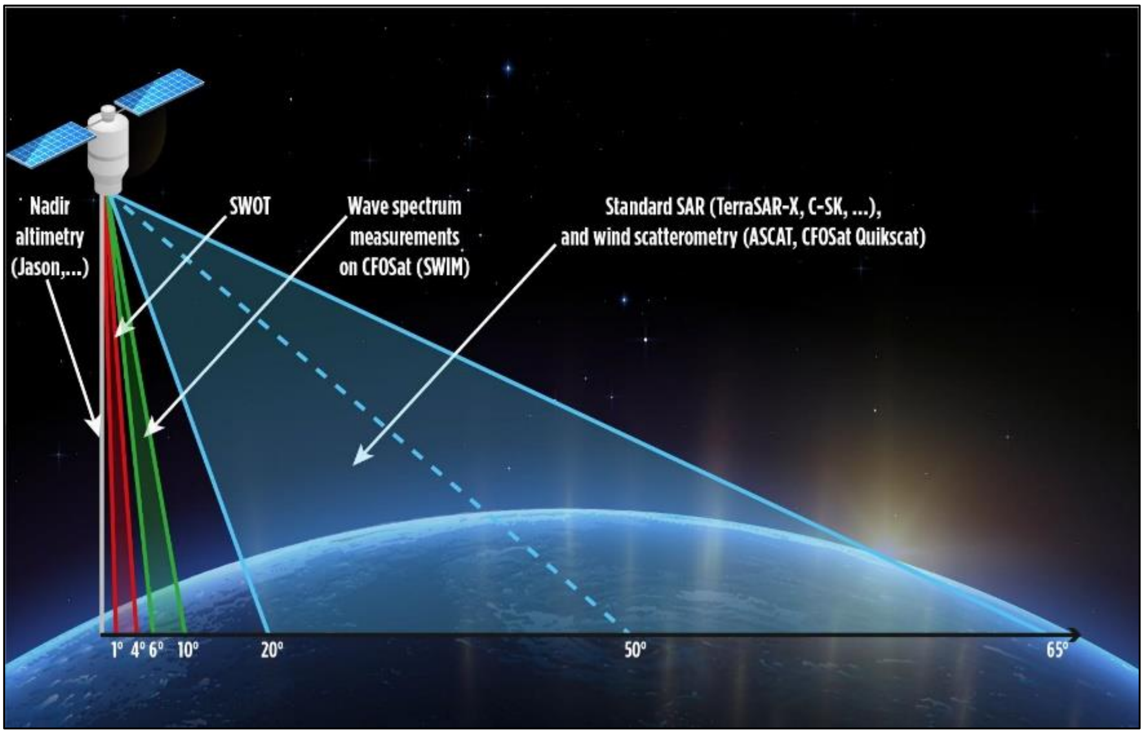

Figure 8. SWOT incidence angle vs other satellites/techniques © CNES/Mira Production

3. Wide- Swath Altimetry

Wide-swath altimetry is an advanced remote sensing technique that measures water surface elevation over a broad, two-dimensional area, rather than just along a narrow track directly beneath a satellite (nadir altimetry). This approach enables the simultaneous observation of a wide "swath" of the Earth's surface, capturing detailed spatial variations in sea surface height (SSH) and inland water bodies.

The measurement relies on radar interferometry, often using SAR, to achieve high spatial resolution and coverage. SWOT mission represents a major milestone in wide-swath altimetry.

3.1 Configuration of SWOT mission

The main mission goals are to improve the spatiotemporal coverage with respect to today's oceanographic radar altimeters (while keeping a height precision of the order of a cm on a km scale grid), and extend the altimetric measurements to continental water surfaces, including lakes, reservoirs, wetlands, and rivers down to a width of 50-100 m (represented on a triangular irregular network with an average spacing of 50 m).

Its satellite includes two kind of altimeters: a dual-frequency (C and Ku-band) conventional altimeter which is only available for the nadir point over a ground footprint of 5-10 km.

The second altimeter is the core instrument: A Ka-band radar Interferometer (KaRin). KaRin contains two Ka-band SAR antennae at opposite ends of a 10-meter boom which forms the interferometric baseline. (Figure 9)

The interferometric is a dual-swath system, alternatively illuminating the left and right swaths on each side of the nadir track. These two swaths measure 50 km wide in a 1°-4° plan.

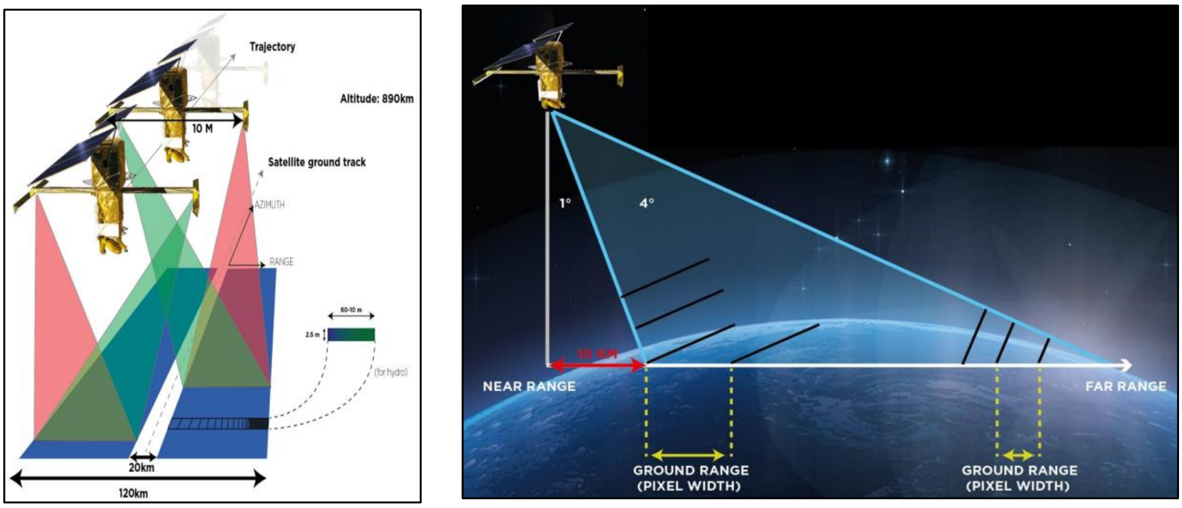

Figure 9. SWOT swath(left) and pixel geometry(right) © CNES/Mira Production.

During its science orbit, SWOT retraces the same path every 21 days, during 292 orbits of the Earth. Each orbit includes an ascending (flying north) and descending (flying south) portion, or "pass". Each pass is numbered (1 to 584). For ascending passes, the time starts near the South Pole. For descending orbits, the time starts near the North Pole.

The ability to do swath altimetry (2D) is such a major advancement with respect to 1D profiles used for 25 years, that it might be the most important technological breakthrough in satellite altimetry since TOPEX/Poseidon's capacity to reach centimetre-level accuracy. [3]

3.1.1 Fundamentals of SWOT Measurements

The SWOT mission collects primary data sets by measuring backscatter and phase. Backscatter evaluates the energy returned from radar pulses, helping estimate water locations, while phase involves timing measurements for assessing water surface elevation using differences in signal return between antennas. (Figure 10)

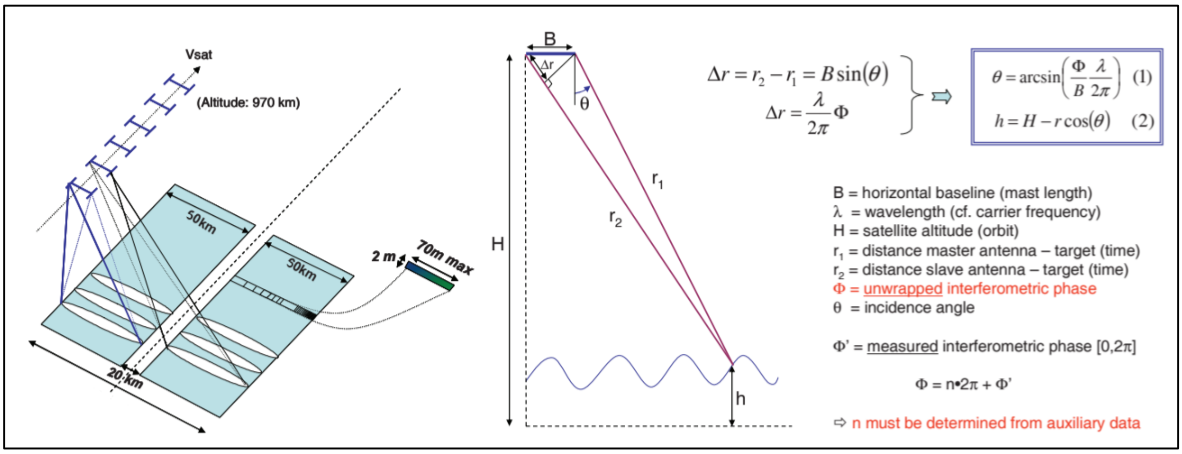

Figure 10. Illustration of the acquisition geometry of KaRIn on SWOT (left) and measurement principle (right). [5]

The principal instrument KaRIn (Ka-band Radar Interferometer) is a bistatic SAR system operating in Ka-band, covering two near nadir swaths (incidence angles 1-4°) on both sides of the satellite track. The interferometric measurement concept is basically triangulation where the baseline B forms the base. H represents the altitude of the satellite relative to an arbitrary reference surface and is measured by precise location system onboard (Doris, GPS/GNSS), h is the surface height that we want to determine (relative to the same reference surface). The phase difference Φ enables us to estimate very precisely the view angle θ.

In terms of processing and data products, KaRIn features a low resolution (LR) mode dedicated to oceanography (km scale grid) and a high resolution (HR) mode dedicated mainly to continental hydrology (triangular irregular network with 50 m average spacing). One of the key features of the KaRIn configuration is its near-nadir incidence. Any terrain feature presenting a slope greater than 1° in near range and 4 ° in far range will produce a layover, disturbing the final image analysis. As we are close to nadir, the backscattering coefficient of water is generally assumed to be much higher than that of land surfaces, in which case the layover could have limited impact. [5]

Click this link https://swot.jpl.nasa.gov/resources/142/swot-global-coverage/ to visualise how KaRIn will collect data.

4. Details of Data Availability

4.1 Radar Altimeter

4.1.1 DAHITI

The Database for Hydrological Time Series of Inland Waters (DAHITI) is published by the Deutsches Geodätisches Forschungsinstitut der Technischen Universität München (DGFI-TUM) and provides water level time series derived from satellite radar altimetry since 2013. DAHITI integrates data from multiple satellite missions such as Envisat, ERS-2, Jason-1/2, and SARAL/AltiKa. Data can be accessed at https://dahiti.dgfi.tum.de/en/.

4.1.2 Hydroweb

Hydroweb is managed by LEGOS (Laboratoire d'Etudes en Géophysique et Océanographie Spatiales), part of CNES/IRD/CNRS/Université de Toulouse, and has been providing inland water level time series since the early 2000s. It compiles data from a range of radar altimeter satellites, including TOPEX/Poseidon, Jason series, Envisat, ERS, and SARAL/AltiKa. Data are available at https://hydroweb.theia-land.fr/.

4.1.3 VEDAS

VEDAS (Visualisation of Earth observation Data and Archival System) is an initiative by the Indian Space Research Organisation (ISRO), offering access to satellite-derived hydrological data since 2013, particularly from the SARAL/AltiKa mission and other Indian satellites. The platform can be accessed at https://vedas.sac.gov.in/hydro.

4.1.4 AVISO

AVISO (Archiving, Validation and Interpretation of Satellite Oceanographic data) is operated by Collecte Localisation Satellites (CLS) for the French Space Agency (CNES), providing radar altimetry data since 1992, beginning with TOPEX/Poseidon. AVISO aggregates data from a wide range of missions, including Jason-1/2/3, Envisat, ERS, SARAL/AltiKa, and Sentinel-3. Data access is available at https://www.aviso.altimetry.fr/.https://www.aviso.altimetry.fr/en/missions/current-missions/swot/access-to-data.html

4.2 SWOT Mission Data

4.2.1 SWOT in Space and Time

From its orbit 554 miles (891 kilometres) above the planet, SWOT scans Earth's surface between 78° North and 78° South latitude. The satellite covers the entire planet every 21 days (one cycle), and during that time observes more than 90% of Earth's surface water. [6]

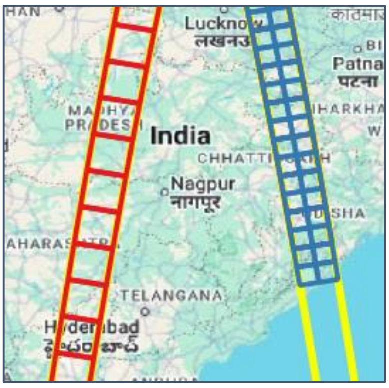

Figure 11 This graphic provides an overview of SWOT's three spatial extent formats: swaths (in yellow), tiles (in blue), and scenes (in red).

It captures data from two 50-kilometer wide swaths on both sides of the spacecraft. The altimeter aboard SWOT collects data, in part, in the 20-kilometer gap between KaRIn's swaths. (Figure 11)

4.2.2 Data Product Types

SWOT HR Product (used for hydrololgy) are described in Figure 12 and different product type are also mentioned below (Figure 13 and 14)

Data Product	Data Product Short Name	Spatial Extent Format	Description	Select Variables	Data Format
Water Mass Pixel Cloud	SWOT_L2_HR_PIXC_2.0	Tile (64x64 km2)	Point cloud of water mask pixels ("pixel cloud").	Geolocated heights, back-scatter, geophysical fields, and flags	NetCDF
Pixel Cloud Vector Attribute	SWOT_L2_HR_PIXCVec_2.0	Tile (64x64 km2)	Auxiliary info for pixel cloud product indicating water bodies pixels are assigned.	Height-constrained pixel geolocation after reach- or lake-scale averaging	NetCDF
Raster	SWOT_L2_HR_Raster_2.0	Scene (128x128 km2)	Geographically fixed rasterized water surface elevation and inundation extent.	Water surface elevation, area, water fraction, backscatter, geophysical information	NetCDF
River Vector	SWOT_L2_HR_RiverSP_2.0	Continent-scale swath	Vectors of river reaches (~10 km long) and nodes (~200 m spacing) in prior river database.	Water surface elevation, slope, width, derived discharge*	Shapefile
Lake Vector	SWOT_L2_HR_LakeSP_2.0	Continent-scale swath	Vectors of lakes in prior lake database and detected features not in the prior river or lake databases.	Water surface elevation, area, derived storage change	Shapefile
Cycle Average River Vector	SWOT_L2_HR_RiverAvg_2.0		Cycle average and aggregation of river reach pass data within predefined hydrological basins.		Shapefile
Cycle Average Lake Vector	SWOT_L2_HR_LakeAvg_2.0		Cycle average and aggregation of lake pass data within predefined hydrological basins.		Shapefile
Floodplain Digital Elevation Model	SWOT_L2_HR_FPDEM*		Flood Plain Digital Elevation Map in raster format, derived from multiple cycles of SWOT acquisitions. (~50m resolution). Provides height and quality flag for each pixel.		NetCDF

*Available ~2 years after launch

Figure 12. Level 2 High Rate (HR) Products. (NASA, 2024)

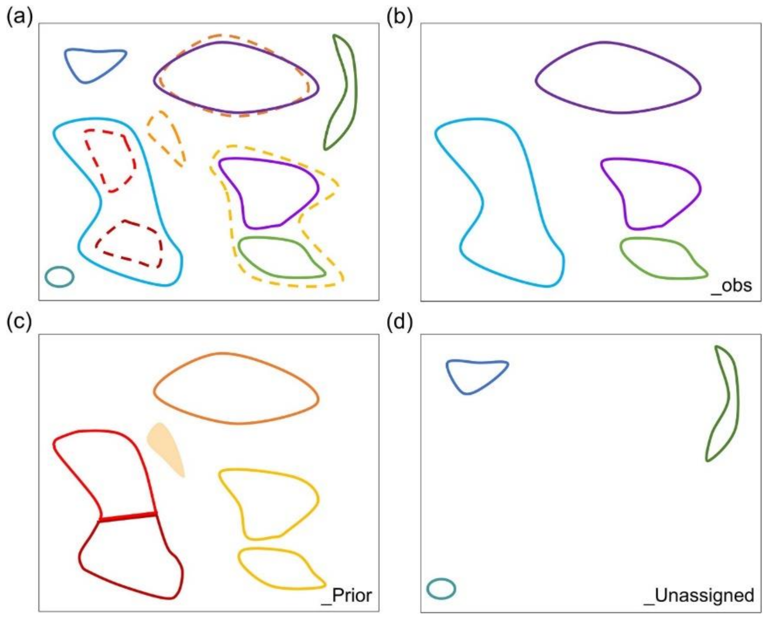

Figure 13. Schematic illustration of how the SWOT mission Prior Lake Database (PLD) is used to organize SWOT observed water features into the three vector files of the L2_HR_LakeSP product. (a) Observed water features (solid) and prior lakes (dashed) in a hypothetical region. Different colors represent different water features or prior lakes. (b) Result of the observation oriented file (L2_HR_LakeSP_Obs). (c) Result of the PLD oriented file (L2_HR_LakeSP_Prior). The unobserved prior lake is an empty geometry with only prior attributes, shown as a filled polygon. An observed feature intersecting two prior lakes is partitioned to two features (red and dark red), whereas two observed features intersecting the same prior lake (yellow) are dissolved to a multipart feature. (d) Result of the observation oriented unassigned file (L2_HR_LakeSP_Unassigned). [16]

The SWOT satellite will survey rivers wider than 100 m (328 ft). Three products will be delivered to users. They will only be generated for rivers recorded in an existing database and will cover reaches of approximately 10 km (6 mi).

The first of these is a **reach product**. On each SWOT satellite overpass, observations are attached to the predefined reaches of 10km each stored in the database. This product provides the mean height, width, slope and discharge of the river for the reach in question. (Figure 14). The second is a **node product**. A node is defined every 200 m (656 ft) along the center line of the river. This product provides the river height and width at each node. Finally, the third is a **cycle average product**. As with lakes, a river may be observed once, twice, three or four times, and even up to 12 times at high latitudes over each 21-day cycle. For each river section, the product compiles all data collected over the full cycle. It's currently estimated that SWOT will observe around 200,000 river reaches across the globe. [17]

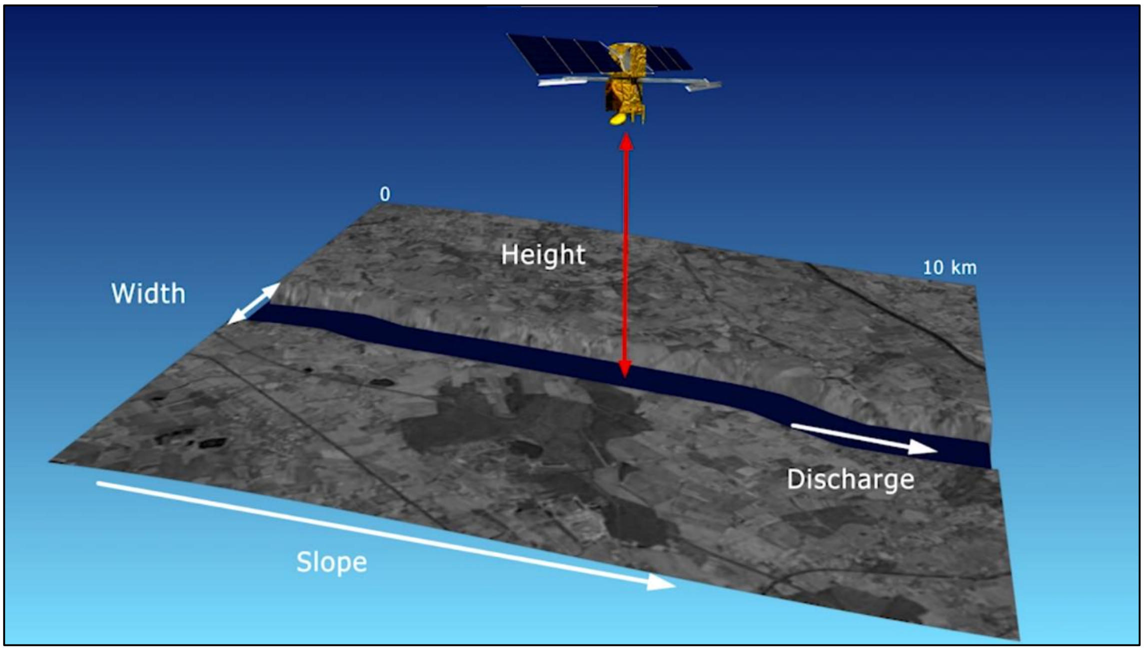

Figure 14. Illustration of the SWOT Reach Product Type (JPL, NASA 2019)

4.2.3 Data Access

4.2.3.1 Hydroweb

SWOT data can be accessed through Hydroweb, managed by LEGOS/CNES/IRD/CNRS/Université de Toulouse, which provides selected SWOT data products at https://hydroweb.theia-land.fr/

4.2.3.2 Earthdata(PoDAAC)

NASA's Physical Oceanography Distributed Active Archive Center (PoDAAC) at JPL, which offers global access to official SWOT mission products via https://podaac.earthdata.nasa.gov/.

4.2.3.3 AVISO

AVISO (Archiving, Validation and Interpretation of Satellite Oceanographic data) is operated by Collecte Localisation Satellites (CLS) for the French Space Agency (CNES), SWOT Data access is available at https://www.aviso.altimetry.fr/en/missions/current-missions/swot/access-to-data.html.

5. Case Studies

5.1 Application of radar altimetry for inland water bodies

5.1.1 A framework for generating rating curves in Mahanadi River using hydrodynamic model and radar altimetry data

The limited availability of in situ data has drawn attention towards using remote sensing techniques to monitor river flow. Radar altimetry data has been used to generate stage–discharge rating curves through power-law relations and empirical methods. However, evaluating hydrodynamic models for rating curve generation using multi-mission altimetry data in data-scarce regions is lacking. We used altimetry data (Jason 2, Jason 3, SARAL/AltiKa, Sentinel 3A, and Sentinel 3B) over the Mahanadi River to evaluate rating curves at virtual stations (Figure 16). The Hydrologic Engineering Center's River Analysis System (HEC-RAS) hydrodynamic model was set up, identifying seven virtual stations along the Mahanadi River from Boudh to Mundali Barrage. (Figure 15) Statistical evaluation of "water level" (Root Mean Square Error [RMSE] 0.27–0.88 m) and "discharge" (Kling-Gupta Efficiency [KGE] 0.52–0.88) components of the generated rating curves showed significant agreement with radar altimetry data. These rating curves at virtual stations offer a cost-effective tool for monitoring river flows at additional locations. [8]

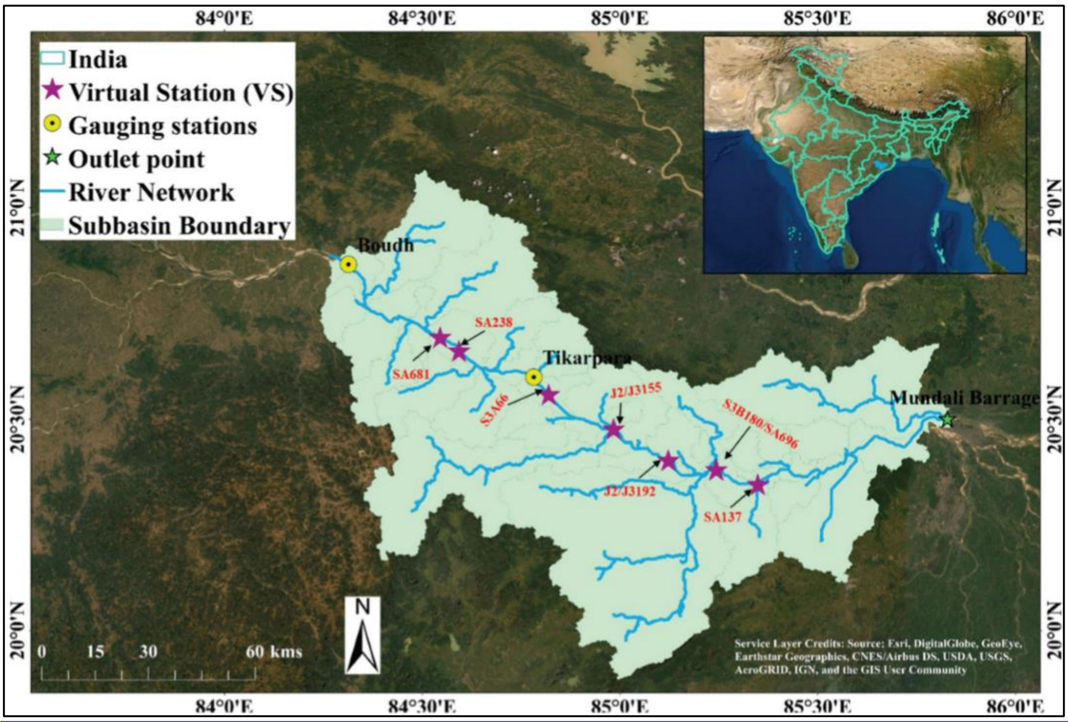

Figure 15. Study Area for Case Study 5.1.1 [8]

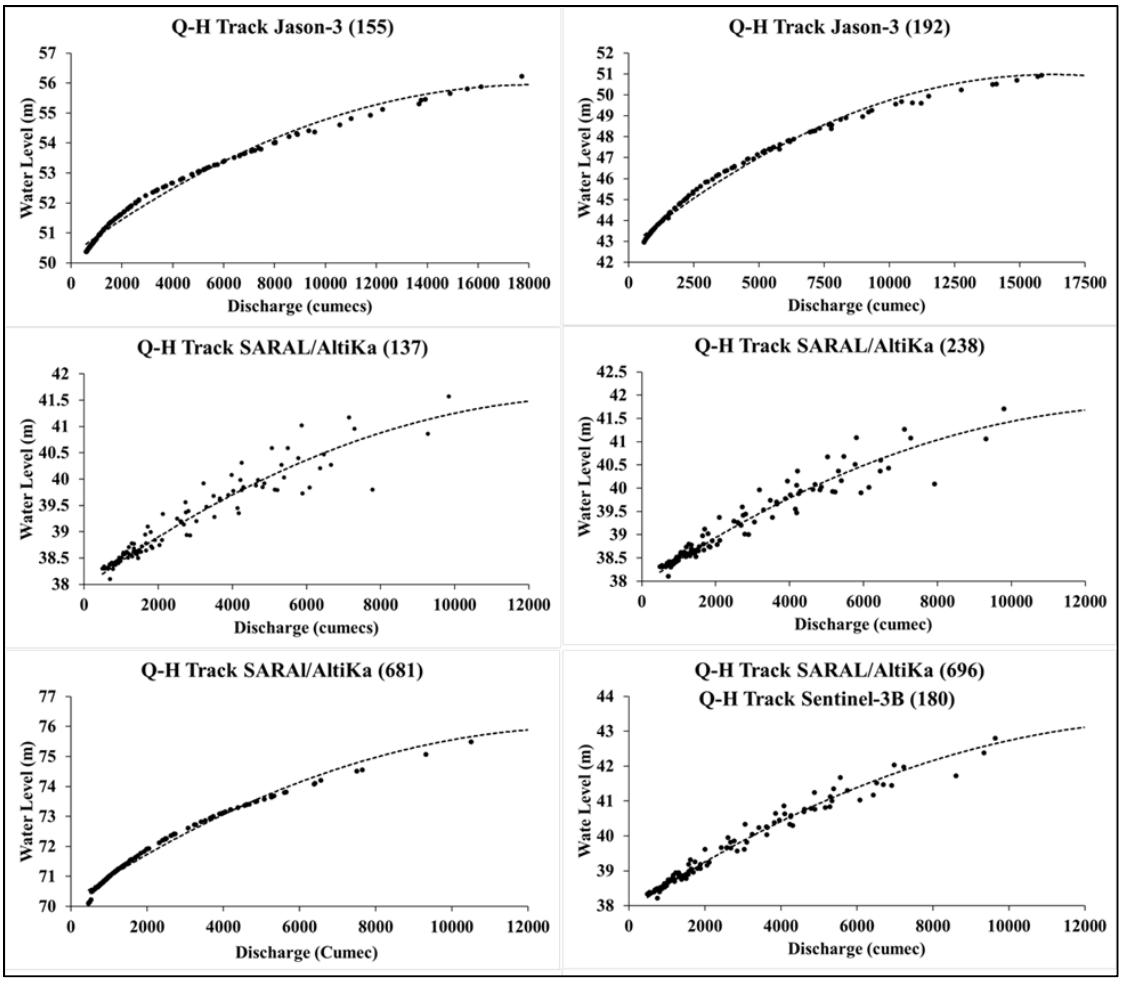

Figure 16 . Stage-discharge rating curves generated at various virtual stations. [8]

5.1.2 Water level status of Indian reservoirs: A synoptic view from altimeter observations

Most of the part of India is already under water-stressed condition. In this regard, the continuous monitoring of the water levels (WL) and storage capacity of reservoirs, lakes, and rivers is very important for the estimation and utilization of water resources effectively. The long term ground observed WL of many of the water bodies is not easily available, which may be very critical for proper water resources management.

Satellite radar altimetry is the remote sensing technique, which is being used to study sea surface height for the last three decades. The advancement in radar technology with time has provided the opportunity to exploit the technique to retrieve the WL of inland water bodies. In the current study, an attempt has been made to generate long term time series on WL of around 29 geometrically complicated inland water bodies in India. The WL of these water bodies was retrieved for around two decades using the European Remote-Sensing Satellite – 2 (ERS-2), ENVISAT Radar Altimeter – 2 (ENVISAT RA-2), and SARAL -AltiKa altimeters data through Ice-1 retracking algorithm. Further, an attempt has also been made to estimate the WL of gauged/ungauged lakes namely Mansarovar, Pangong, Chilika, Bhopal, and Rann of Kutch over which SARAL-AltiKa pass was there. As after July 2016, the SARAL-AltiKa is operating in the drifting orbit, systematic repeated observation of WL data of all reservoirs was not possible. The data of drifted tracks of SARAL -AltiKa were tested for WL estimation of Ban Sagar reservoir. As the ERS-2, ENVISAT RA-2 and SARAL -AltiKa all were having almost the same passing tracks, a long term WL series of these lakes could be generated from 1997 to 2016.

However, at present only Sentinel – 3 is in orbit, the continuous altimeter based WL monitoring of some of these reservoirs (Gandhi Sagar, Nathsagar, Ranapratap, Ujjani, and Ukai) was attempted through Sentinel-3A satellite data from 2016 to 2018. (Figure 17) The accuracy of the retrieved WL was then validated against the observed WL. In most of the reservoirs, a systematic bias was found due to the different characteristics and geoid height of each reservoir. The coefficient of determination, R^2, value for a majority of reservoirs was as good as 0.9. In the case of ERS-2, the values of R^2 varied for 0.44–0.97 with root mean square error (RMSE) in the range of 0.63–2.72 m. These statistics improved with the ENVISAT RA-2 data analysis, the R^2 value reached more than 0.90 for around 11 reservoirs. The highest, 0.99, for Hasdeo and Shivaji Sagar Reservoirs with RMSE of 0.44 and 0.56, respectively. Further, the accuracy improved with the analysis of SARAL -AltiKa data. The R^2 was always more than 0.9 for each reservoir and the lowest RMSE reduced to 0.03. Therefore, it can be said that the accuracy and consistency of WL retrieval through satellite altimetry has improved with time. Furthermore, the altimeter based retrieved WL may be used in hydrological studies and can contribute to better water resources management. [7]

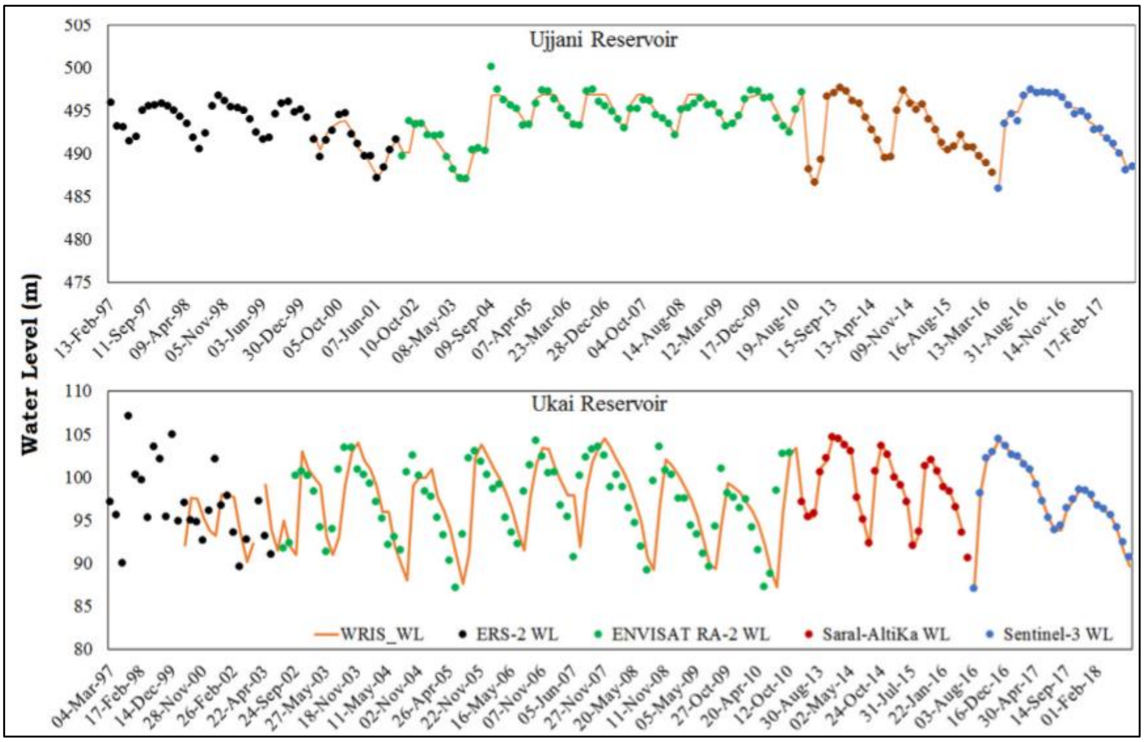

Figure 17. Continuous WL time series derived with various altimeter data. [7]

5.2 Application of wide-swath altimetry for inland water bodies

The SWOT mission represents a major advancement in global hydrology by providing detailed, two-dimensional water height data across various inland water bodies. This capability enhances hydrological models by improving the estimation of river discharge, water storage changes, and fluxes within the water cycle. The integration of SWOT observations with in-situ measurements and other satellite datasets will refine our understanding of surface-subsurface water interactions and the effects of climate change on hydrological systems. Furthermore, SWOT's data can contribute to broader Earth system studies, including research on greenhouse gas emissions from inland waters, ultimately improving climate models and informing sustainable water management strategies. The study was done at various locations across India including rivers, reservoirs, weir, hydro-power project and glacial lakes (Figure 19). Here, we illustrate water level variations upstream (U/S) and downstream (D/S) of the Pulichintala Hydro-Power Project on the Krishna River, located along the border of Telangana and Andhra Pradesh.

The visualisation is from the SWOT Raster Product (Level-2 HR Raster - 100m) and the value extraction for the time series is from the SWOT Node Product (Level-2 HR River SP - Node). The time series analysis reveals that water levels peak during the monsoon months (a significant 30m dip from U/S to D/S) and gradually decline as water is diverted for irrigation and other uses.

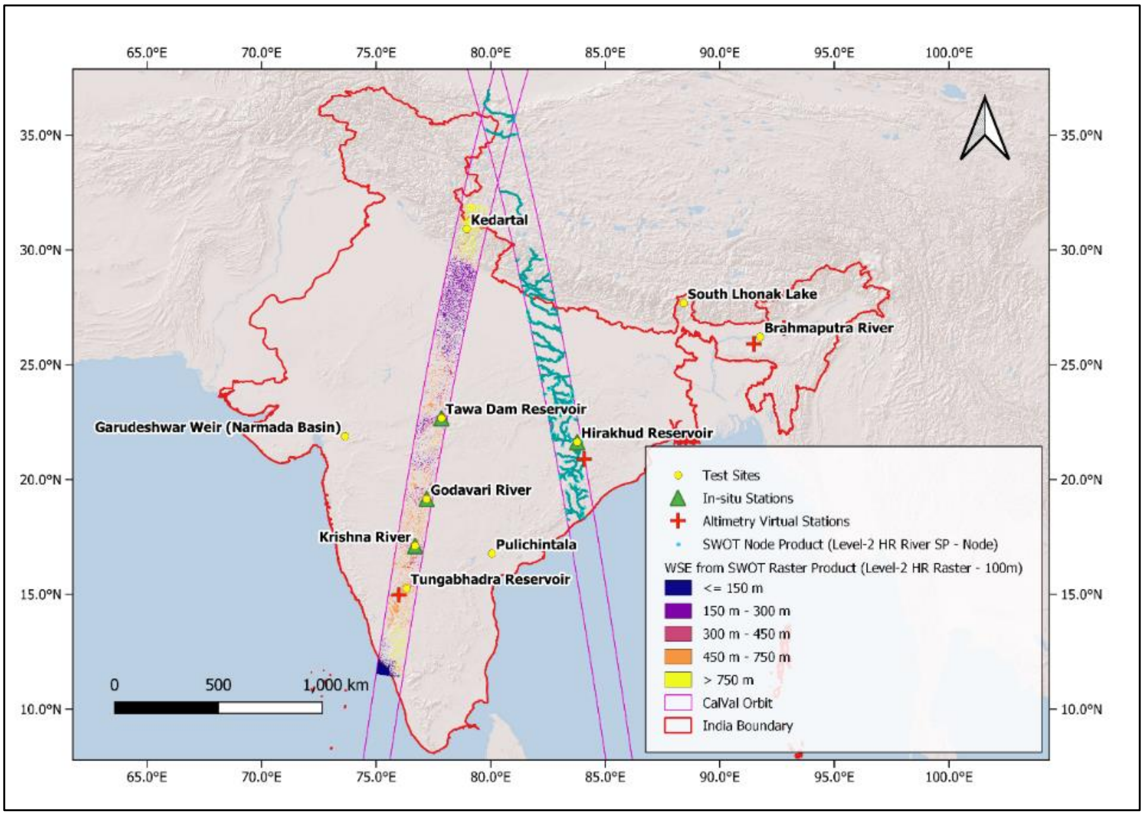

Figure 18. Study Area for Case Study 5.2

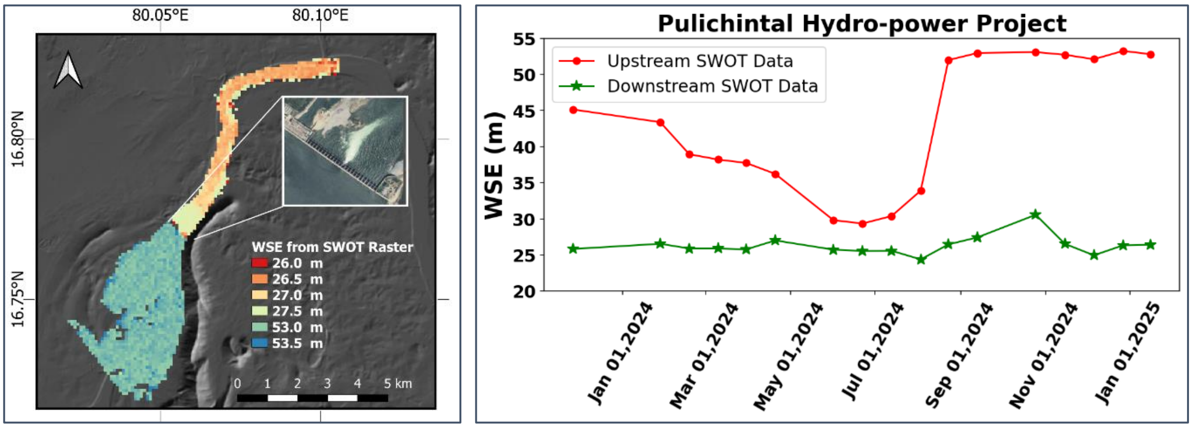

Figure 19. SWOT HR Raster Product Visualisation (left) and timeseries for U/S and D/S (right) for Pulichintala Hydropower project.

6. Conclusion

The advent of advanced satellite altimetry, culminating in the SWOT mission, marks a transformative era in the monitoring and management of Earth's surface water resources. By leveraging cutting-edge radar interferometry and wide-swath technology, SWOT provides unprecedented high-resolution, two-dimensional observations of global oceans, lakes, and rivers. This capability fills critical gaps left by traditional ground-based and nadir-pointing satellite observations, enabling more accurate assessments of water levels, streamflow, and the dynamics of inland water bodies. The data from SWOT not only enhances our understanding of the global water cycle but also supports improved hydrological modelling, river discharge prediction, and water resource management—vital for addressing the challenges posed by climate change, population growth, and increasing water demands.

7. Challenges ahead

Despite its significant advancements, the SWOT mission and satellite altimetry in general face several limitations and challenges:

- Spatial and Temporal Resolution Constraints - While SWOT offers improved coverage, its revisit time (21 days) may not capture rapid hydrological events such as flash floods or short-term water level fluctuations. Smaller water bodies or narrow rivers may still fall below the detection threshold or be affected by mixed pixel issues.

- Atmospheric and Environmental Interference - Signal accuracy can be compromised by atmospheric conditions (e.g., heavy rainfall, ionospheric disturbances) and surface roughness, especially in vegetated or urbanized regions.

- Calibration and Validation - Ground truth data for calibration and validation remain sparse in many regions, limiting the accuracy and reliability of satellite-derived measurements.

- Accessibility and Usability - The complexity and volume of SWOT data may pose challenges for end-users, particularly in developing regions with limited technical capacity.

8. Future Prospective

The future of satellite altimetry and missions like SWOT is bright, with several promising directions:

- Higher Temporal and Spatial Resolution - Future missions may offer more frequent revisits and finer spatial resolution, enabling near-real-time monitoring of dynamic water systems.

- Integration with Other Data Sources - Combining SWOT data with in-situ sensors, other satellite missions (e.g., optical, thermal, LIDAR), and advanced modelling will provide a more holistic and actionable view of the water cycle.

- Broader Applications - SWOT data will support a wide range of applications, from disaster risk reduction (flood and drought monitoring), ecosystem management, and transboundary water governance to climate change adaptation and urban planning.

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